

Is exposure to cyanobacteria an environmental risk factor for amyotrophic lateral sclerosis and other neurodegenerative diseases?

There is a broad scientific consensus that amyotrophic lateral sclerosis (ALS) is caused by gene-environment interactions. Mutations in genes underlying familial ALS (fALS) have been discovered in only 5-10% of the total population of ALS patients. Relatively little attention has been paid to environmental and lifestyle factors that may trigger the cascade of motor neuron death leading to the syndrome of ALS, although exposure to chemicals including lead and pesticides, and to agricultural environments, smoking, certain sports, and trauma have all been identified with an increased risk of ALS. There is a need for research to quantify the relative roles of each of the identified risk factors for ALS. Recent evidence has strengthened the theory that chronic environmental exposure to the neurotoxic amino acid β -N-methylamino-L-alanine (BMAA) produced by cyanobacteria may be an environmental risk factor for ALS. Here we describe methods that may be used to assess exposure to cyanobacteria, and hence potentially to BMAA, namely an epidemiologic questionnaire and direct and indirect methods for estimating the cyanobacterial load in ecosystems. Rigorous epidemiologic studies could determine the risks associated with exposure to cyanobacteria, and if combined with genetic analysis of ALS cases and controls could reveal etiologically important gene-environment interactions in genetically vulnerable individuals.